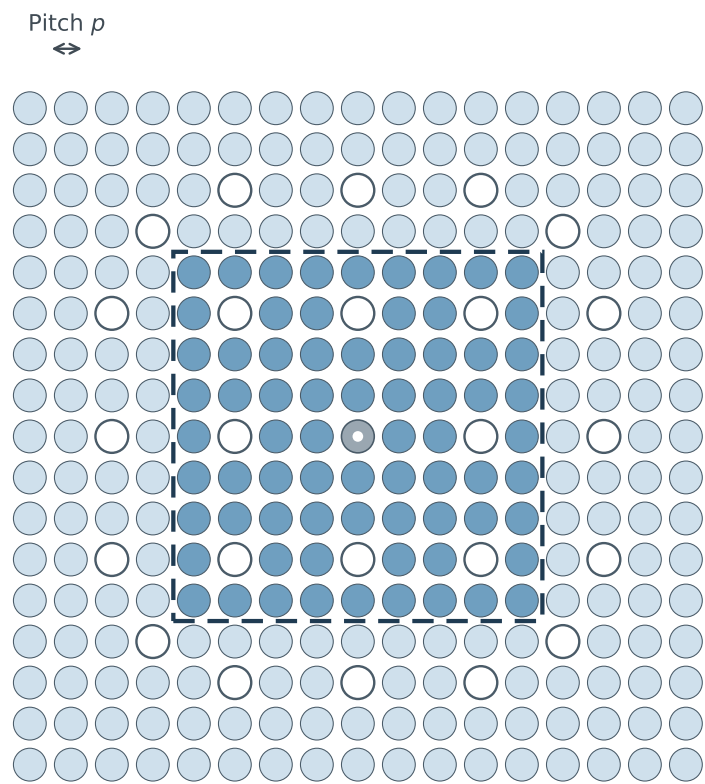
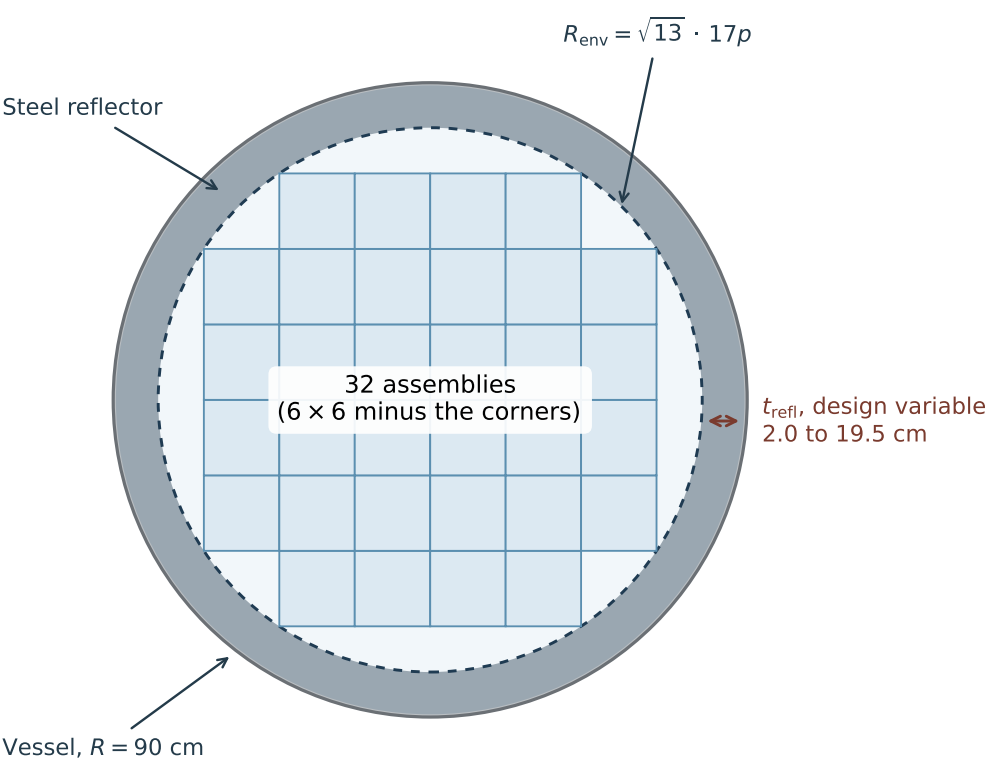


(a) The 17×17 assembly



- Inner zone, e_{in} (72 rods)
- Guide tube (24)
- Outer zone, e_{out} (192 rods)
- Instrument tube (1)

(b) The core plan view, drawn at $p = 1.26$ cm and $t_{refl} = 12$ cm



$g_{geom} = R_{env}(p) + \delta_{pad} + t_{refl} - 90 \leq 0, \quad \delta_{pad} = 0.02 \text{ cm}$

Radial budget at this drawing: $R_{env} = 77.23$ cm, reflector outer = 89.23 cm, slack = 0.75 cm

The six design variables

e_{in}	Inner-zone enrichment	2.00 to 19.75	wt% ^{235}U	p	Lattice pitch	1.15 to 1.43	cm
e_{out}	Outer-zone enrichment	2.00 to 19.75	wt% ^{235}U	t_{refl}	Reflector thickness	2.0 to 19.5	cm
w_{Gd}	Gadolinia weight fraction	0.0 to 8.0	wt% Gd_2O_3	n_{Gd}	Gadolinia rod count	12 to 40	rods

264 fuel rods per assembly, 72 in the inner zone and 192 in the outer zone. p and t_{refl} are coupled through g_{geom} , so the ranges above bound a box and not a feasible region.